Building a High-Availability Architecture in Elasticsearch

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Why High Availability Matters

In mission-critical environments, downtime isn't just inconvenient—it's costly. Elasticsearch powers real-time search and analytics for enterprises worldwide.

Continuous Operation

Ensures service stays online despite node or infrastructure failures

Data Protection

Prevents data loss in critical search and analytics workloads

User Experience

Supports business continuity with minimal disruption

Elasticsearch HA: Three Key Levels of Resilience

Index Level

Shard replication protects data and improves query performance

Node Level

Multiple nodes eliminate single points of failure

Zone Level

Distribute nodes across availability zones for fault tolerance

How Do You Run Elasticsearch?

Self-Managed (Cloud / On-Prem)

- Full control over architecture
- Required for some regulations
- **HA is your responsibility**

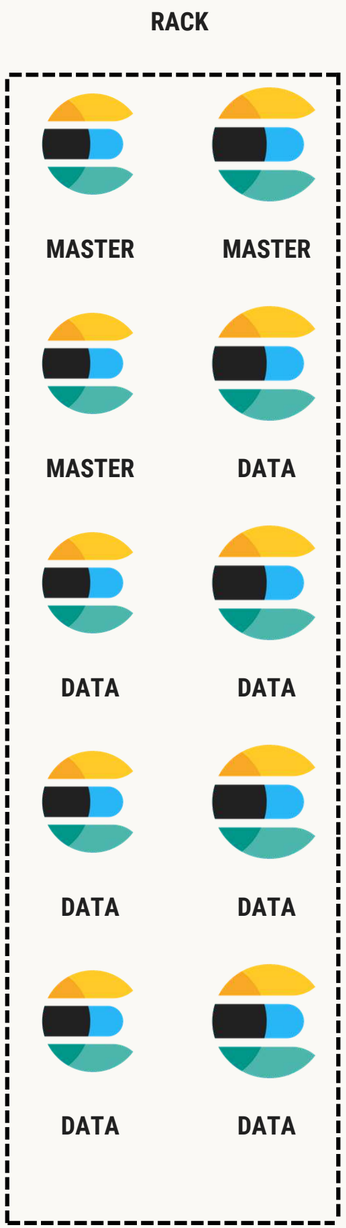
Elastic Cloud Hosted

- Managed by Elastic
- More configuration control
- Zone awareness handled by platform

Elastic Cloud Serverless

- Fully managed
- Minimal operational control
- Best for small or focused use cases

Source: Elastic blog



Where Do We Start Designing for HA?

01

Initial Setup

You build a 10-node cluster. Everything works perfectly.

02

The Problem

Your entire rack fails. Your entire cluster goes down.

03

The Question

How can we prevent this catastrophic failure?

Physical Separation Strategy

Single Rack

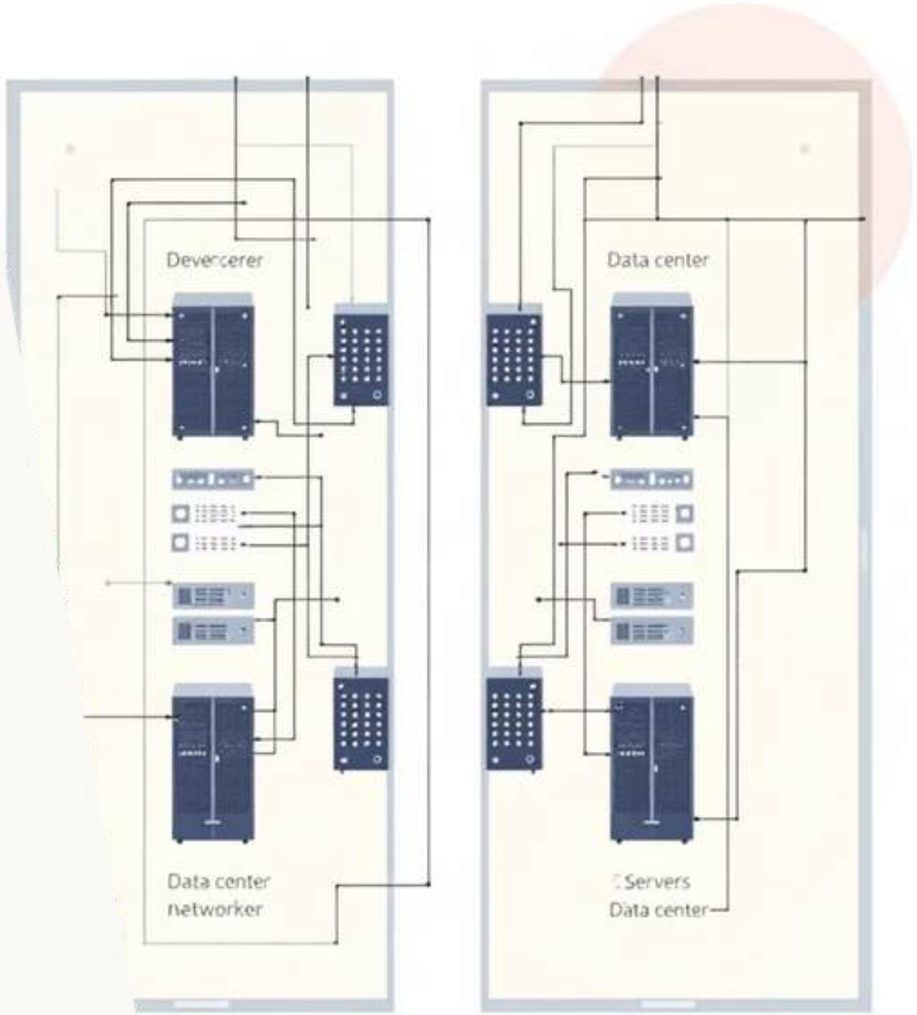
All nodes in one location—high risk

Two Server Rooms

Same region, different rooms—balanced approach

Different Regions ?

Maximum availability but the problem is latency + coordination



Shard Allocation Awareness

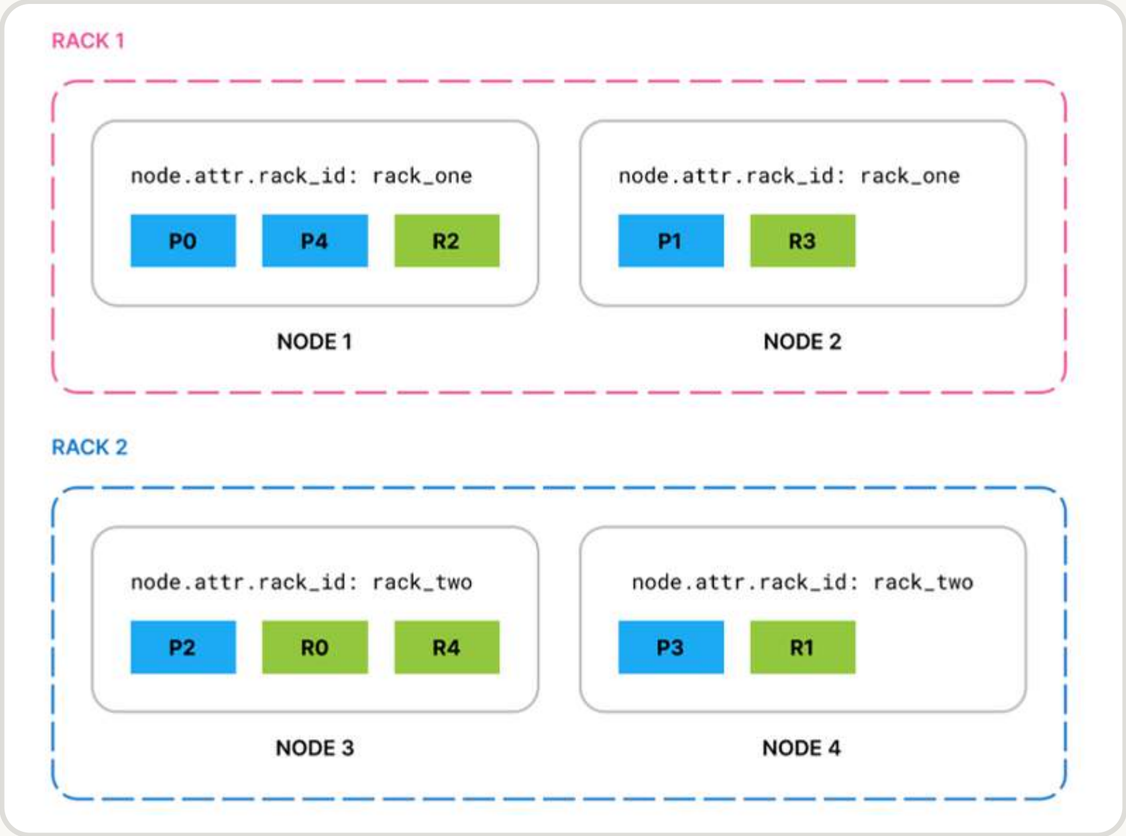
Having nodes in two rooms isn't enough—you need proper shard distribution.

The Challenge

Are your shards distributed equally with backups on both sides? Probably not unless you configured shard allocation awareness.

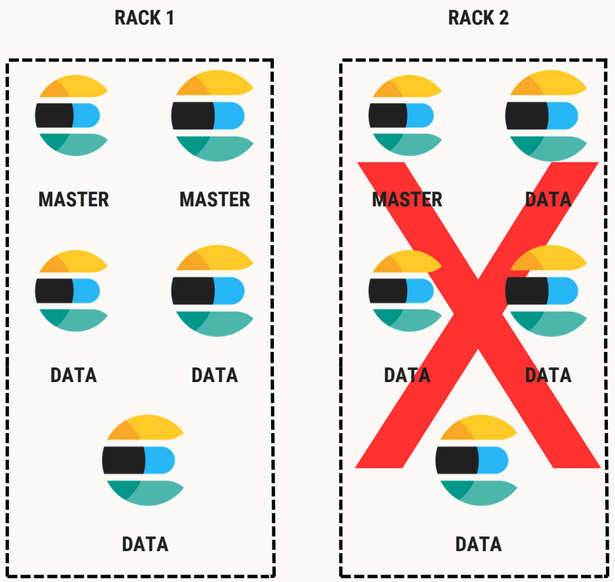
What It Does

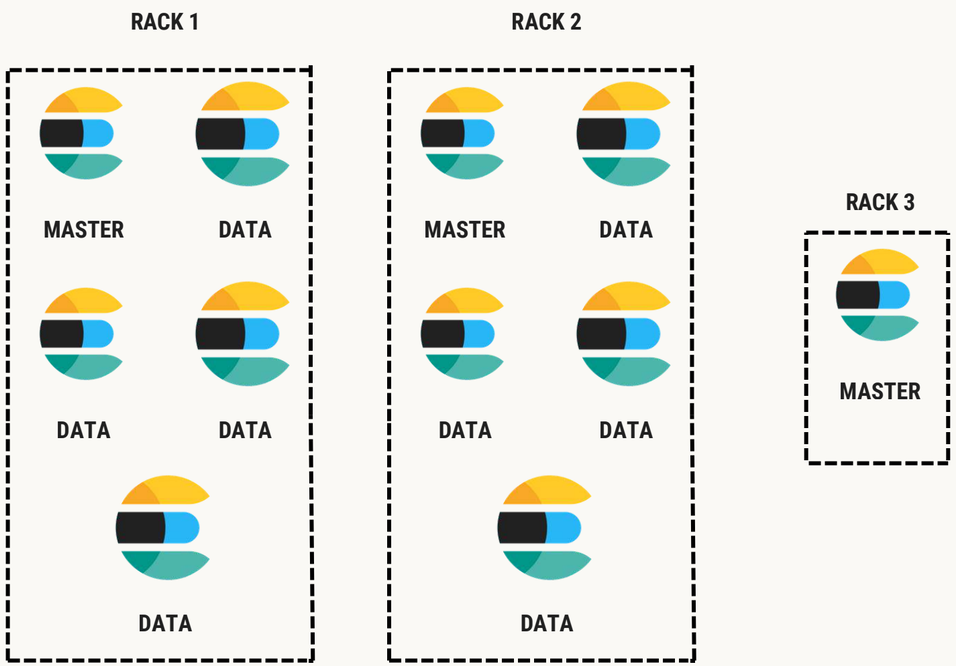
Elasticsearch considers your physical hardware configuration when allocating shards, distributing primary and replica shards to minimize risk.



Source: Elastic documentation

Master Node Configuration





Best Practice

- Distribute master nodes across server rooms
- Add a third master-eligible node in an independent zone

Snapshot Strategy (Backup & Recovery)

1. **Configure Snapshots**
   Repository + retention policy

2. **Schedule Wisely**
   Avoid peak indexing and query load

3. **Test Recovery**
   Verify recovery time and data integrity

Critical Warning: Snapshots consume IO and network bandwidth. Poor timing or overlap with heavy indexing can impact cluster performance.

Searchable Snapshots

Elasticsearch's searchable snapshots transform backups into an integral part of your data architecture, enabling cost-effective long-term data retention.

Beyond Backup

Directly query data stored in snapshots, turning them into an active part of your operational data store, not just a recovery mechanism.

Enabling Frozen Tiers

Facilitates the frozen data tier, allowing older, less frequently accessed data to be moved from expensive local disks to more affordable object storage.

Cost-Effective Retention

Store vast amounts of historical data for extended periods in solutions like AWS S3, Azure Blob, or Google Cloud Storage, while still retaining query capabilities.

Data Tiers & ILM: Scaling Data Without Breaking HA

Hot Tier

Frequently accessed data requiring fast performance

Warm Tier

Less frequently queried, optimized for storage

Cold Tier

Rarely accessed, cost-effective storage

Frozen Tier

Archival data, accessed 1-2 times per year

ILM automates data movement across tiers — design tiers with failure domains in mind.

Load Balancing Strategy

Basic Approach

Connect applications directly to multiple Elasticsearch nodes — acceptable for small clusters, but hard to manage at scale.

Recommended Solution

Implement a dedicated load balancer to:

- Distribute requests evenly
- Route to specific node types (hot nodes, coordinating-only nodes)
- Simplify application configuration
- Isolate client traffic from cluster topology changes

Load balancers eliminate the headache of managing individual node connections and ensure optimal resource utilization.

Cross-Cluster Replication for Disaster Recovery

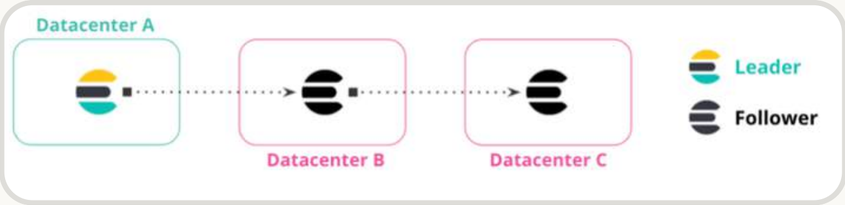
Designed for regional failures and full-cluster outages. Cross-cluster replication helps maintain service continuity during major failures.

Single DR Datacenter

Active-passive model with one follower cluster in different region

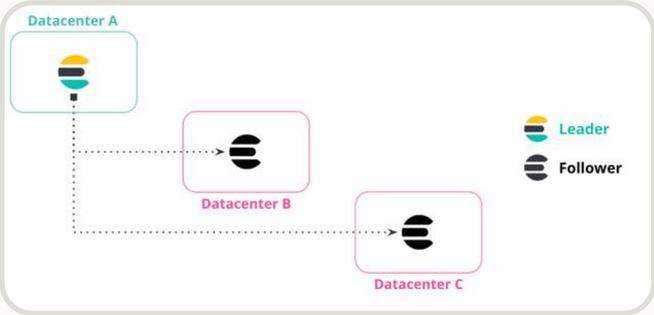
Chained Replication

Data flows through multiple datacenters in sequence



Multiple DR Datacenters

One leader cluster replicating to multiple follower clusters



Bi-directional Replication

Read and write are supported in multiple clusters

Limitation: CCR replicates user indices only. System indices, security configuration, and cluster state are not replicated.

All Images from Elastic documentation

THANK YOU

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